

Problem Set 1

Applied Stats/Quant Methods 1
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Answer Sheet

Due: October 9, 2025

Question 1: Education

Part 1 - 90% confidence interval for the average student IQ in the school:

Code in R:

```
1 y <- c(105, 69, 86, 100, 82, 111, 104, 110, 87, 108, 87, 90, 94, 113,  
        112, 98, 80, 97, 95, 111, 114, 89, 95, 126, 98)  
2 #part 1  
3 mean_iq <- mean(y) #calculating the mean  
4 sd_iq <- sd(y) #calculating the standard deviation  
5 se_iq <- sd_iq/sqrt(length(y)) #calculating the standard error  
6 alpha <- (1-0.9)/2 #because is is 90% CI  
7 tscore <- qt(alpha, df = length(y)-1, lower.tail = F) #calculating t  
        score for n<30 because it is 25  
8 lower_90 <- mean_iq - (tscore * se_iq) #lower CI interval  
9 upper_90 <- mean_iq + (tscore * se_iq) #upper CI interval  
10 confint90 <- c(lower_90, upper_90)  
11 confint90
```

Execution answer: The confidence interval is

Explanation: Since the length of y is 25, it is less than 30. Hence, it is not a normal distribution. It should be t-distribution. I started with calculating the mean, then the standard deviation, then the standard error using the standard deviation. Then I calculated the t-score and using the t-score, I calculated the upper and lower interval to get the confidence interval.

Part 2 - Conducting the appropriate hypothesis test with $\alpha = 0.05$

Code in R:

```
1 Ho <- 100 #we want to prove that average IQ in school greater than 100 –  
  alternative Hypothesis, so our null hypothesis is that it is 100  
2 alpha2 <- 0.05 #given alpha  
3 test_statistic = (mean_iq-Ho)/se_iq #test statistic is the difference  
  between means with the standard error as denominator  
4 p_value <- pt(test_statistic, length(y)-1, lower.tail = F) #should be 1  
  sided  
5 p_value #0.2784617 which is >0.05 hence we fail to reject Null hypothesis  
  – this means that there is not enough evidence to conclude that the  
  average IQ in this school is higher than 100
```

Execution answer: The p value is 0.2784617 which is >0.05 (alpha) hence we fail to reject Null hypothesis - this means that there is not enough evidence to conclude that the average IQ in this school is higher than 100.

Explanation: I set the null hypothesis is equal to 100 while the alternative hypothesis is that the average IQ is greater than 100. I used the regular formula for the test statistic and used it to calculate the p value with degrees of freedom $\text{length}(y) - 1$. This is 1 sided test because the alternative is that it is higher and not "not equal to".

Question 2: Political Economy

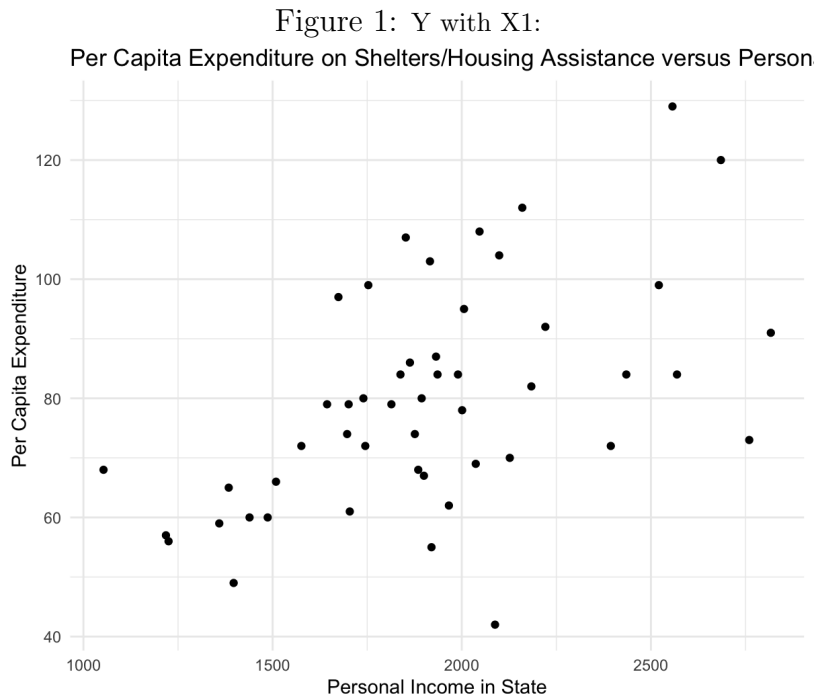
Part 1 - Correlation plots between Y, X1, X2, and X3:

Plot 1 - Y with X1 - Personal income versus shelter/housing expenditure assistance:

Code in R:

```
1 #Y with X1 - Personal income versus shelter/housing expenditure
  assistance
2 ggplot(expenditure, aes(x = X1, y = Y)) +
3   geom_point() +
4   labs(
5     title = "Per Capita Expenditure on Shelters/Housing Assistance versus
      Personal Income",
6     x = "Personal Income in State", y = "Per Capita Expenditure") + theme
      _minimal()
7 cor(expenditure$X1, expenditure$Y) #the correlation is postive and
  upwards but not very strong
```

Explanation: The correlation is postive and upwards but not very strong, but it looks linear. This is a scatter plot that shows the relationship between the personal income and the shelter/housing expenditure assistance. Below is the plot:

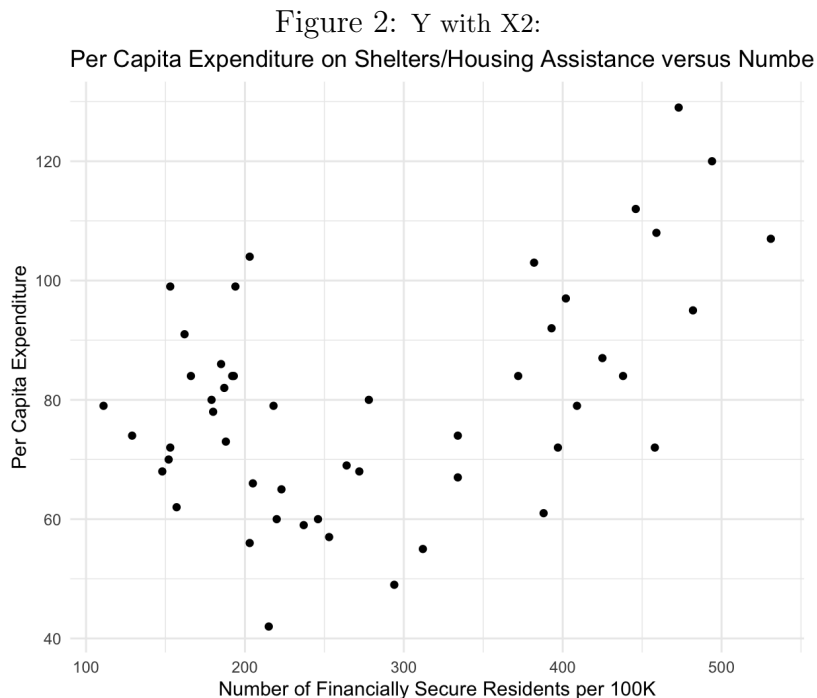


Plot 2 - Y with X2- Per Capita Expenditure on Shelters/Housing Assistance versus Number of Financially Secure residents:

Code in R:

```
1 #Y with X2 – Per Capita Expenditure on Shelters/Housing Assistance versus  
  Number of Financially Secure residents  
2 ggplot(expenditure , aes(x = X2, y = Y)) +  
3   geom_point() +  
4   labs(  
5     title = "Per Capita Expenditure on Shelters/Housing Assistance versus  
  Number of Financially Secure residents",  
6     x = "Number of Financially Secure Residents per 100K", y = "Per  
  Capita Expenditure") + theme_minimal()  
7 cor(expenditure$X2, expenditure$Y) #the correlation is downward then  
  upwards, seems like it is weak
```

Explanation: The correlation is downward then upwards, seems like it is weak, which is non-linear. This is a scatter plot that shows the relationship between the per Capita Expenditure on Shelters/Housing Assistance and Number of Financially Secure residents. Below is the plot:

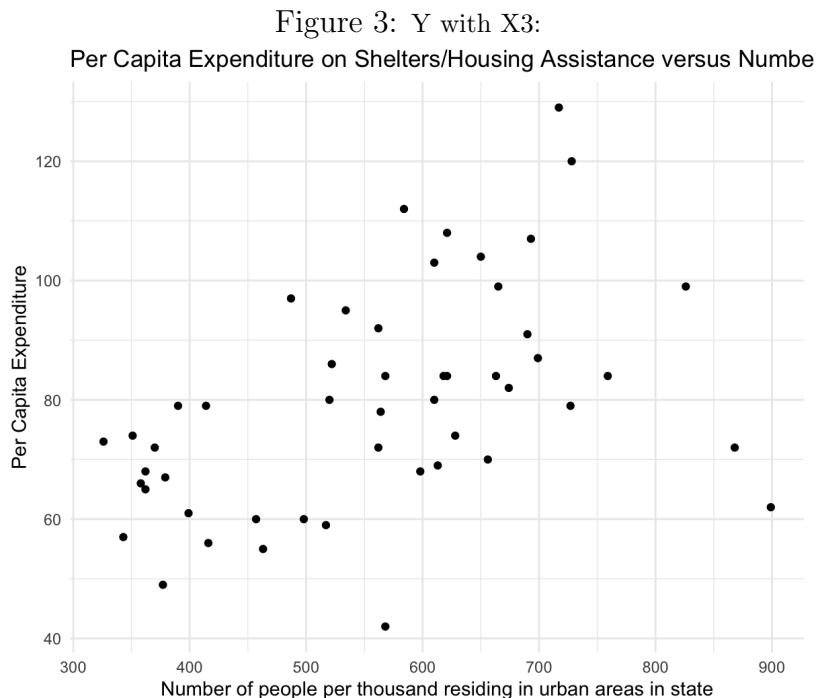


Plot 3 - Y with X3- Per Capita Expenditure on Shelters/Housing Assistance versus Number of people per thousand residing in urban areas in state:

Code in R:

```
1 #Y with X3 – Per Capita Expenditure on Shelters/Housing Assistance versus
  Number of people per thousand residing in urban areas in state
2 ggplot(expenditure, aes(x = X3, y = Y)) +
3   geom_point() +
4   labs(
5     title = "Per Capita Expenditure on Shelters/Housing Assistance versus
  Number of people per thousand residing in urban areas in state",
6     x = "Number of people per thousand residing in urban areas in state",
7     y = "Per Capita Expenditure") + theme_minimal()
8 cor(expenditure$X3, expenditure$Y) #the correlation is upward then
  downward, seems like it is also weak
```

Explanation: The correlation is upward then downward, seems like it is weak, which is non-linear. This is a scatter plot that shows the relationship between the Per Capita Expenditure on Shelters/Housing Assistance and Number of people per thousand residing in urban areas in state. Below is the plot:

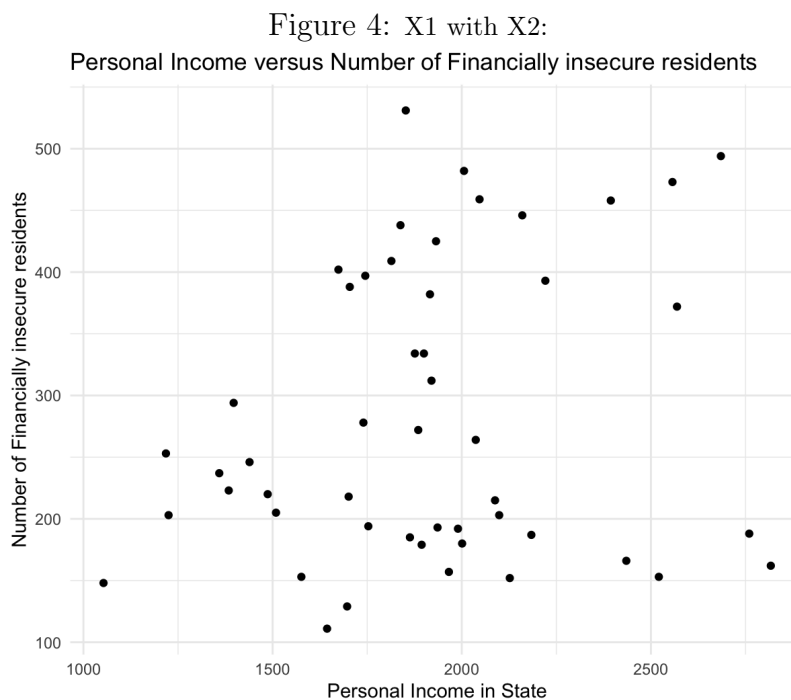


Plot 4 - X1 with X2- Personal Income versus Number of Financially insecure residents:

Code in R:

```
1 #X1 with X2 – Personal Income versus Number of Financially insecure
  residents
2 ggplot(expenditure, aes(x = X1, y = X2)) +
3   geom_point() +
4   labs(
5     title = "Personal Income versus Number of Financially insecure
  residents",
6     x = "Personal Income in State", y = "Number of Financially insecure
  residents") + theme_minimal()
7 cor(expenditure$X1, expenditure$X2) #there is no correlation it seems
```

Explanation: There seems to be no correlation between the Personal Income and Number of Financially insecure residents. Below is the scatter plot:

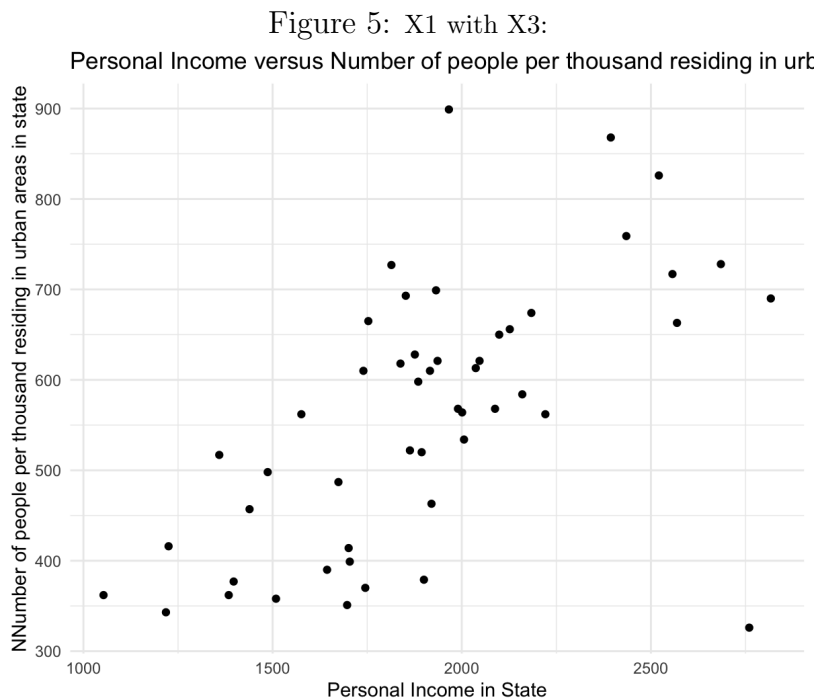


Plot 5 - X1 with X3- Personal Income versus Number of people per thousand residing in urban areas in state:

Code in R:

```
1 #X1 with X3 – Personal Income versus Number of people per thousand
  residing in urban areas in state
2 ggplot(expenditure , aes(x = X1, y = X3)) +
3   geom_point() +
4   labs(
5     title = "Personal Income versus Number of people per thousand
  residing in urban areas in state",
6     x = "Personal Income in State", y = "NNumber of people per thousand
  residing in urban areas in state") + theme_minimal()
7 cor(expenditure$X1, expenditure$X3) #seems the correlation is upward and
  positive
```

Explanation: The correlation between Personal Income and Number of people per thousand residing in urban areas in state seems to be upward and positive. Below is the scatter plot:

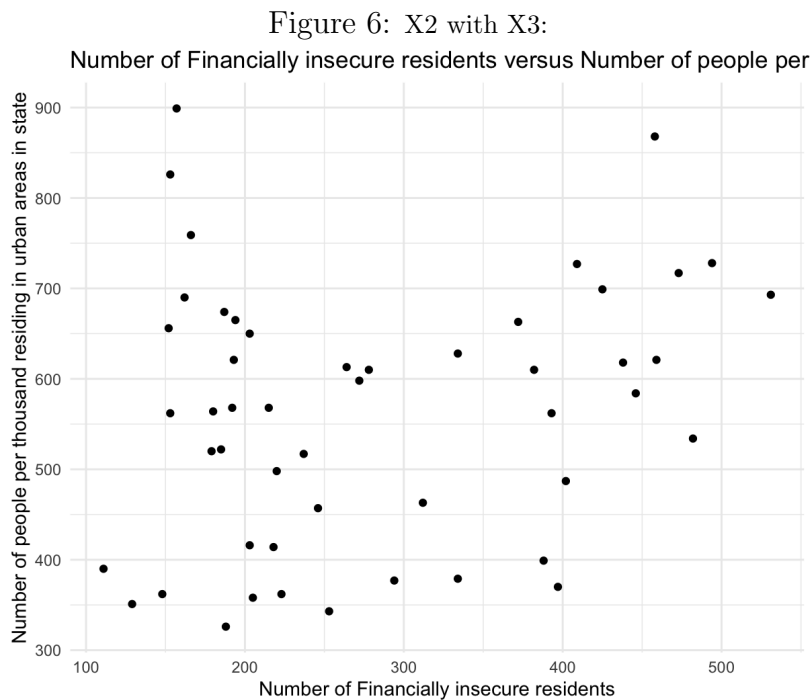


Plot 6 - X2 with X3- Number of Financially insecure residents versus Number of people per thousand residing in urban areas in state:

Code in R:

```
1 #X2 with X3 – Number of Financially insecure residents versus Number of
  people per thousand residing in urban areas in state
2 ggplot(expenditure , aes(x = X2, y = X3)) +
3   geom_point() +
4   labs(
5     title = "Number of Financially insecure residents versus Number of
  people per thousand residing in urban areas in state",
6     x = "Number of Financially insecure residents", y = "Number of people
  per thousand residing in urban areas in state") + theme_minimal()
7 cor(expenditure$X2, expenditure$X3) #there is no correlation it seems
```

Explanation: There seems to be no correlation between the Number of Financially insecure residents and number of people per thousand residing in urban areas in state. Below is the scatter plot:



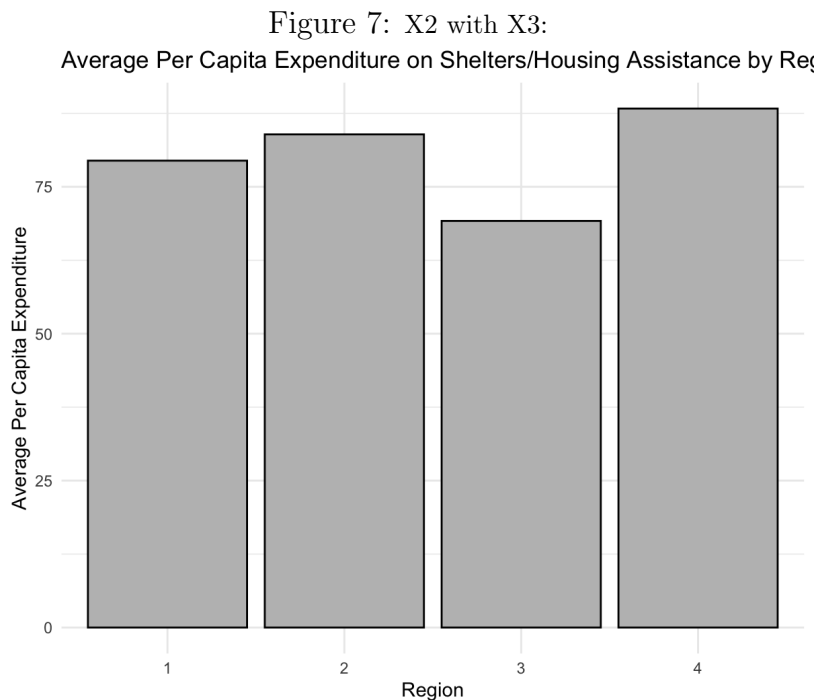
Part 2 - Average capita expenditure by region:

Y with Region - Average Per Capita Expenditure on Shelters/Housing Assistance by Region:

Code in R:

```
1 #Y and Region
2 ggplot(expenditure, aes(x = factor(Region), y = Y)) +
3   geom_bar(stat = "summary", fun = "mean", fill = "grey", color = "black"
4     ) + #using the mean because you want to see the average per region and
5       this function works on Y statistics
6   labs(
7     title = "Average Per Capita Expenditure on Shelters/Housing
8       Assistance by Region",
9     x = "Region", y = "Average Per Capita Expenditure") + theme_minimal()
10 #On average, the West region, region 4, has the highest per capita
11   expenditure on housing assistance.
```

Explanation: On average, the West region, region 4, has the highest per capita expenditure on housing assistance. I used the mean because you want to see the average per region and this function works on Y statistics. Below is the plot:

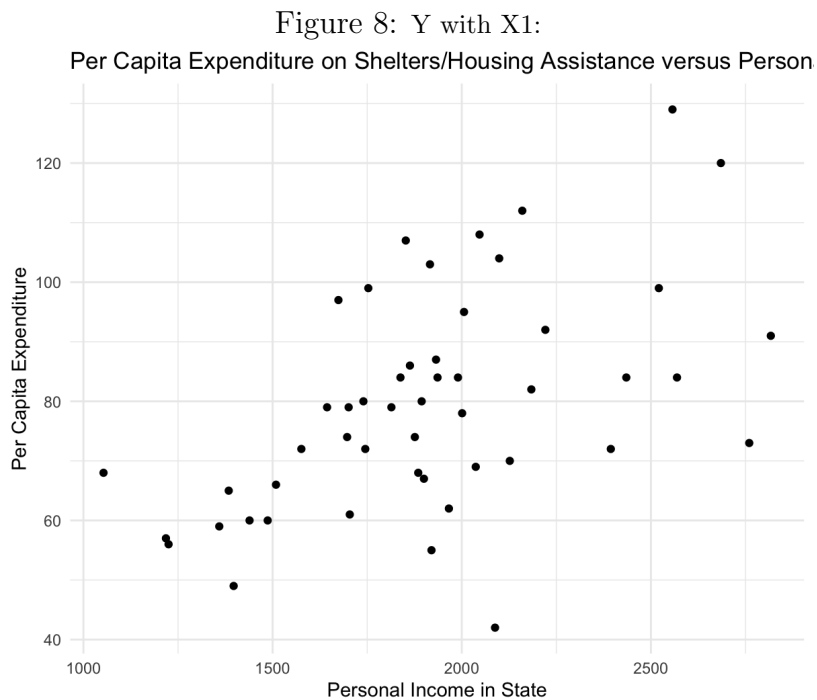


Part 3A - Y with X1 - Personal income versus sheler/housing expenditure assistance:

Code in R:

```
1 #Y with X1 – Personal income versus sheler/housing expenditure assistance
2 ggplot(expenditure , aes(x = X1, y = Y)) +
3   geom_point() +
4   labs(
5     title = "Per Capita Expenditure on Shelters/Housing Assistance versus
6     Personal Income",
7     x = "Personal Income in State", y = "Per Capita Expenditure") + theme
8     _minimal()
```

Explanation: Plotting the relationship between Y and X1. The correlation is downward then upwards, seems like it is weak, which is non-linear. This is a scatter plot that shows the relationship between the per Capita Expenditure on Shelters/Housing Assistance and Number of Financially Secure residents. It seems the relationship is not very strong. Below is the plot:



Part 3B - Y with X1 and Region - Per Capita Expenditure on Shelters/Housing Assistance versus Personal Income by Region:

Code in R:

```
1 #Y with X1 with Region
2 ggplot(expenditure, aes(x = X1, y = Y, color = factor(Region), shape =
3   factor(Region))) +
4   geom_point() +
5   labs(
6     title = "Per Capita Expenditure on Shelters/Housing Assistance versus
7       Personal Income by Region",
8     x = "Personal Income in State", y = "Per Capita Expenditure", color =
9       "Region", shape = "region") + theme_minimal()
```

Explanation: This plot shows the relationship Per Capita Expenditure on Shelters/Housing Assistance versus Personal Income by Region. The regions are in different characters and shapes. Below is the plot:

